

Course Project: Replication and Extension of a Recent Bayesian Paper

Objective:

This project challenges you to critically engage with contemporary research in Bayesian theory and/or application. You will select a recent (within the last 7 years) peer-reviewed paper involving Bayesian statistical methods, replicate its core results, and extend the work either theoretically or through additional simulation or application.

Project Components

1. Paper Selection:

Select a modern peer-reviewed article on a topic relevant to Bayesian statistics. The paper must include original methodological contributions, theory development, simulation studies, or an applied analysis.

Please plan on providing the chosen paper to the instructor NLT Friday, 17 Apr 2026 @ 1300. *This selection should include a short summary (3-5 sentences) of why the paper you have chosen is feasible to replicate and what extension you are considering.*

Where to Find Papers:

- *Bayesian Analysis* (International Society for Bayesian Analysis)
- *Journal of the American Statistical Association* (especially Theory & Methods or Applications sections)
- *Journal of the Royal Statistical Society Series B*
- *Annals of Statistics*
- *Technometrics* (for applied Bayesian methods)
- *Statistical Science* (for conceptual or review-style Bayesian papers)
- *arXiv* – Statistics (stat) or Machine Learning (cs.LG) sections
- *Google Scholar* (search “Bayesian hierarchical model,” “Bayesian MCMC,” “posterior inference,” etc.)
- *CRAN* (look for packages with published papers or vignettes describing Bayesian methods)

Tip: Look for papers that include simulation studies, posterior summaries, or applied examples that you can replicate (e.g., using MCMC or conjugate models). Strong project papers will clearly describe the model, provide enough detail to reproduce results, and often suggest limitations or extensions you can build on.

2. Initial Presentation (Week 6):

You'll give a 7-10 minute in-class presentation that introduces:

- The problem being addressed in the paper
- The methodology used
- Key results or figures to be replicated
- Your plan for extending the work

This is meant to get early feedback from both the instructor and peers and to solidify your project direction.

3. Replication and Extension Work:

You will replicate the key findings from the paper. This might include:

- Running or re-creating simulation studies
- Verifying theoretical results
- Re-analyzing datasets using the paper's methods

Then, extend the paper in a meaningful way. Examples include:

- Generalizing the model or relaxing an assumption
- Adding a new simulation condition or comparison
- Applying the method to a novel dataset
- Proposing a new estimator or reliability function

4. Final Presentation (Week 10 or Finals Week):

Deliver a formal 15–20 minute presentation covering:

- Summary of the original paper and its contributions
- Your replication process and findings
- The extension you carried out and your results
- Reflection on challenges, insights, and what you learned

5. Final Report (Due Finals Week):

Submit a technical report (approx. 8–12 pages) including:

- Abstract and Introduction (problem background, paper overview)
- Replication Methods and Results
- Extension Approach and Findings

- Discussion (limitations, future directions)
- References

You may include code in an appendix or submit as a separate file.

Suggestions for Success

- Start early and select a paper with methods or results you feel confident analyzing.
- Document everything — code, plots, tables — even things that didn't work as expected.
- Be creative with your extension. Try something you're curious about.
- Ask questions. I'm happy to help clarify the original paper, debug code, or talk through ideas.